CS 450: Get OpenCode working, step by step

Wednesday, October 21, 2026. Today's goal: OpenCode running on your account, and one small checked task done with it. Matt leads the steps from the front. Go at the room's pace, and don't skip ahead.

> How to use this sheet
> Every step has three parts: Type (exactly this), You should see (something like this), and If not (which fix to use, at the back of the sheet).
> Your key is yours. Never paste it into a chat, a screenshot or a message, and never type it where others can see it. Nobody else types it or reads it out: not your neighbour, not an AI tool.
> Stuck? Find your fix number. Not listed? Raise your hand. Working already? Help your neighbour.

Step 0: Did you set this up before? Check first (everyone, 3 minutes)

Don't redo anything yet. Redoing it can overwrite a setup that already works.

0.1 Did you get OpenCode working on agate or your laptop on Monday, or ever?

- Never tried, or no idea: skip to Pick your route.
- Tried on agate: log in (Route A, steps A1–A2), then do 0.2.
- Tried on your laptop: open Terminal (Mac) or PowerShell (Windows), then do 0.2.

0.2 Test it in a fresh, empty folder. Type one line at a time:

```
cd ~
mkdir cs450-oct21-a
cd cs450-oct21-a
ls
git status
```

You should see: `ls` prints nothing (the folder is empty), and `git status` says

```
fatal: not a git repository (or any of the parent directories): .git
```

Windows: `mkdir` prints a small table first; that's fine. `git` not found or not recognized? Also fine. `mkdir` says the folder exists? Type `cd ~` and use `cs450-oct21-a-2` instead. `git status` shows `On branch …`? You're inside a project folder: stop and ask Matt.

Now start OpenCode:

```
opencode
```

OpenCode opens a full-screen view with a box to type in. First choose the course model: type `/models`, press Enter, and pick DeepThought Claude Sonnet 4.6. Then type:

```
Create hello.py that prints Hello CS 450, run it, and show the output.
```

If it asks permission to create or run the file, allow it.

You should see, in OpenCode's answer

```
Hello CS 450
```

…and `hello.py` is in the folder, made with DeepThought Claude Sonnet 4.6 selected. Then you're done: go to Part 2 (in this folder), and help a neighbour. Quit OpenCode first by typing `/exit`.
If not: first quit OpenCode if it opened (`/exit`) and type `cd ~` to go back to your home folder. Then: `command not found` → start your route at A4 (agate) or B1 (laptop). Any other error → on agate do 0.3; on a laptop start at B3.

0.3 agate only: Monday's setup. Monday's handout may have left a key line in `.bashrc`. Check without showing anything (this prints a number of matching lines, never the key):

```
grep -c DEEPTHOUGHT_API_KEY ~/.bashrc
```

You should see

```
0      (no Monday key line: go on)
1      (or more: there may be a Monday key line)
```

If 1 or more: lock the file so other agate users can't read it. It prints nothing:

```
chmod 600 ~/.bashrc
```

Then check whether your settings file is Monday's handout file. Again this prints a number only:

```
grep -c '{env:DEEPTHOUGHT_API_KEY}' ~/.config/opencode/opencode.json
```

```
1      (Monday's handout file: move it aside with the line below)
0      (a different file: don't touch it; ask Matt at checkpoint A)
… No such file or directory      (no file: go on)
```

Only if it printed 1, move it aside. This line refuses to overwrite an earlier backup:

```
[ -e ~/.config/opencode/opencode-monday.json ] && echo "backup already exists: ask Matt" || mv ~/.config/opencode/opencode.json
~/.config/opencode/opencode-monday.json
```

It prints nothing when it worked. Then start Route A at step A3. Never open `.bashrc` on the projector, and never `echo` the key.

Pick your route (1 minute)

Route	Pick it if
A. agate (first choice)	You have an agate account. Any laptop works, Mac or Windows.
B. your laptop	Your agate login fails twice (F1–F4 didn't fix it), or you have no agate account.
Neither	No laptop, a Chromebook or a tablet: sit with a neighbour and do Part 2 on their screen.

Route A: agate, step by step

Open Terminal (Mac: Cmd+Space, type Terminal) or PowerShell (Windows: Start, type PowerShell).

A1. Connect

Type, with your own UNH username before the `@` (same on Mac and Windows):

```
ssh your-unh-username@agate.cs.unh.edu
```

You should see (first time only)

```
The authenticity of host 'agate.cs.unh.edu (…)' can't be established.
…
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

Type `yes` and press Enter. If not: `Connection timed out` → F4.

A2. Password

You should see

```
your-unh-username@agate.cs.unh.edu's password:
```

Type your UNH password. Nothing appears while you type. That's normal. Press Enter.

You should see a prompt like

```
your-unh-username@agate:~$
```

If not: `Permission denied` → F1. Password won't take → F2. Not on the list → F3.

> ✔ Checkpoint A: your screen shows `your-unh-username@agate:~$`. Neighbour checks: ____

A3. Is OpenCode installed?

```
opencode --version
```

You should see a number starting with 1, like

```
1.19.3
```

Got a number starting with 1? Skip to A5. `opencode: command not found`? Do A4. A number starting with 0 or 2 → F9.

A4. Install OpenCode (type one line at a time, Enter after each)

```
npm config set prefix ~/.npm-global
echo 'export PATH="$HOME/.npm-global/bin:$PATH"' >> ~/.bashrc
source ~/.bashrc
npm install -g opencode-ai@1
```

You should see: the first three lines print nothing. The last one ends with a line like

```
added ⋯ packages in ⋯s
```

Now check again:

```
opencode --version
```

```
1.19.3
```

If not: `command not found` → F5. `EACCES` → F6.

> ✔ Checkpoint B: `opencode --version` prints 1.something. Neighbour checks: ____

A5. Get the course setup helper

```
curl -O https://blkfin.github.io/cs450/setup/opencode/setup-key.py
ls setup-key.py
```

You should see (after a small download table)

```
setup-key.py
```

If not: `No such file or directory` → retype the `curl` line exactly; it is case-sensitive.

A6. Prepare the settings

```
python3 setup-key.py prepare
```

You should see

```
Course configuration ready. No key was read or displayed.
```

If not: `Existing OpenCode configuration differs` → F12. `An alternate OpenCode configuration exists` → F13.

A7. Your key (Matt shows this on screen; wait for him)

Create your Deep Thought key the way Matt shows you, and copy it. Then type:

```
python3 setup-key.py key
```

You should see

```
Paste your DeepThought key (hidden), then press Enter:
```

Paste with a right-click (or Ctrl+Shift+V), and press Enter. Nothing appears when you paste. That's on purpose.

```
Key saved privately. Its value was not printed. You can now open OpenCode.
```

If not: `Key must be nonempty` → F14. `Piped input is refused` → F15.

A8. Check it, without showing the key

```
python3 setup-key.py check
opencode models deepthought
```

You should see

```
Configuration: present
Key file: present
DeepThought connection: not tested
deepthought/ets:aws:us.anthropic.claude-sonnet-4-6
```

If not: `Key file: missing` → redo A7. `Configuration is invalid ⋯ bad file reference` → F17. No model line, or "model not found" → F22.

✔ Checkpoint C: `present`, `present`, and the model line. Neighbour checks: ____

A9. Test it for real, in a fresh folder

```
cd ~
mkdir cs450-oct21-b
cd cs450-oct21-b
ls
git status
```

You should see: `ls` prints nothing (the folder is empty), and `git status` says

```
fatal: not a git repository (or any of the parent directories): .git
```

Windows: `mkdir` prints a small table first; that's fine. `git` not found or not recognized? Also fine. `mkdir` says the folder exists? Type `cd ~` and use `cs450-oct21-b-2` instead. `git status` shows `On branch …`? You're inside a project folder: stop and ask Matt.

Now start OpenCode:

```
opencode
```

OpenCode opens a full-screen view with a box to type in. First choose the course model: type `/models`, press Enter, and pick DeepThought Claude Sonnet 4.6. Then type:

```
Create hello.py that prints Hello CS 450, run it, and show the output.
```

If it asks permission to create or run the file, allow it.

You should see, in OpenCode's answer

```
Hello CS 450
```

If not: it only chats and makes no file → F20. `401` or `403` → F18. It spins for over a minute → F19.

✔ Checkpoint D: with DeepThought Claude Sonnet 4.6 selected, `hello.py` exists in the fresh folder **and** printed `Hello CS 450`. A chat reply alone doesn't count. Matt or neighbour checks: ____ → Go to Part 2.

Route B: your laptop, step by step

Open Terminal (Mac: Cmd+Space, type Terminal) or PowerShell (Windows: Start, type PowerShell). Not Cursor, and not an AI tool's terminal.

B1. Are Node and Python installed?

```
Mac:      node --version
          python3 --version
Windows:  node --version
          py -3 --version
```

You should see two version numbers, like

```
v22.12.0
Python 3.12.4
```

If not: `node` not found → F7. Python not found, or the Microsoft Store opens → F10. Mac asks for "command line developer tools" → F11.

B2. Install OpenCode

```
npm install -g opencode-ai@1
opencode --version
```

You should see a line like `added ⋯ packages`, then

```
1.19.3
```

If not: Mac `EACCES` → F6. Windows "running scripts is disabled" → F8. Windows "is not recognized" → F7.

✔ Checkpoint B: `opencode --version` prints 1.something. Neighbour checks: ____

B3. Get the course setup helper

In your browser, go to blkfin.github.io/cs450/setup/opencode/setup-key.py and save the file (right-click, "Save as"). It lands in Downloads. Check:

```
Mac:      ls ~/Downloads/setup-key.py
Windows:  ls "$HOME\Downloads\setup-key.py"
```

You should see the file listed. If not: F16.

B4. Prepare the settings

```
Mac:      python3 ~/Downloads/setup-key.py prepare
Windows:  py -3 "$HOME\Downloads\setup-key.py" prepare
```

```
Course configuration ready. No key was read or displayed.
```

If not: `Existing OpenCode configuration differs` → F12. `An alternate OpenCode configuration exists` → F13.

B5. Your key (Matt shows this on screen; wait for him)

Create and copy your key the way Matt shows you. Then, the same command with `key` at the end:

```
Mac:      python3 ~/Downloads/setup-key.py key
Windows:  py -3 "$HOME\Downloads\setup-key.py" key
```

```
Paste your DeepThought key (hidden), then press Enter:
```

Paste (Cmd+V on Mac, Ctrl+V or right-click on Windows) and press Enter. Nothing appears.

```
Key saved privately. Its value was not printed. You can now open OpenCode.
```

If not: `Key must be nonempty` → F14. Windows `key folder already exists` → F21. `Piped input is refused` → F15.

B6. Check it, without showing the key

```
Mac:      python3 ~/Downloads/setup-key.py check
Windows:  py -3 "$HOME\Downloads\setup-key.py" check
Both:     opencode models deepthought
```

```
Configuration: present
Key file: present
DeepThought connection: not tested
deepthought/ets:aws:us.anthropic.claude-sonnet-4-6
```

If not: `Key file: missing` → redo B5. `Configuration is invalid ⋯ bad file reference` → F17. No model line, or "model not found" → F22.

✔ Checkpoint C: `present`, `present`, and the model line. Neighbour checks: ____

B7. Test it for real, in a fresh folder

```
cd ~
mkdir cs450-oct21-b
cd cs450-oct21-b
ls
git status
```

You should see: `ls` prints nothing (the folder is empty), and `git status` says

```
fatal: not a git repository (or any of the parent directories): .git
```

Windows: `mkdir` prints a small table first; that's fine. `git` not found or not recognized? Also fine. `mkdir` says the folder exists? Type `cd ~` and use `cs450-oct21-b-2` instead. `git status` shows `On branch …`? You're inside a project folder: stop and ask Matt.

Now start OpenCode:

```
opencode
```

OpenCode opens a full-screen view with a box to type in. First choose the course model: type `/models`, press Enter, and pick DeepThought Claude Sonnet 4.6. Then type:

```
Create hello.py that prints Hello CS 450, run it, and show the output.
```

If it asks permission to create or run the file, allow it.

You should see, in OpenCode's answer

```
Hello CS 450
```

If not: only a chat reply → F20. `401` / `403` → F18. Spins over a minute → F19.

> ✔ Checkpoint D: with DeepThought Claude Sonnet 4.6 selected, `hello.py` exists in the fresh folder **and** printed `Hello CS 450`. A chat reply alone doesn't count. Matt or neighbour checks: ____

Part 2: Your first checked task (minutes 35–45)

Write your check first. Count by hand before the AI does. Count each word in the list as a whole word only, ignoring capital letters and punctuation.

Letter: *Jordan is an exceptional student and a natural leader. Jordan is warm, kind, and helpful to classmates. Jordan has excellent writing skills and strong research skills. In group projects, Jordan takes the lead and keeps the team on track.*

Words: exceptional, excellent, leader, lead, warm, kind, helpful, skills

My hand count, total hits: ____

Now, in OpenCode (in the fresh folder where `hello.py` worked, with DeepThought Claude Sonnet 4.6 selected), type:

```
Save this letter as letter.txt: "Jordan is an exceptional student and a natural leader. Jordan is warm, kind, and helpful to classmates. Jordan has excellent writing skills and strong research skills. In group projects, Jordan takes the lead and keeps the team on track." Then write count.py that counts how many times each of these words appears in letter.txt as a whole word, ignoring capital letters and punctuation: exceptional, excellent, leader, lead, warm, kind, helpful, skills. Run it and show the count for each word and the total.
```

OpenCode's total: ____ Same as your count? yes / no

If they differ, find the word where they differ. Who was right, you or the AI? Which rule did each of you use?

> ✔ Done when `count.py` exists, it ran, and you checked its total against your own count. Neighbour checks: ____

Part 3: Still stuck at minute 45?

That's fine. Email Matt (mfm2@unh.edu) today with these four lines, so he can fix it with you before Monday:

```
Route: agate / laptop
OS: Mac / Windows (version if you know it)
Last step that worked: (for example A6 or B4)
Failing command and its exact output: (copy and paste the text)
```

Before you send it, delete any key or password from what you pasted. Text beats a screenshot.

In-person help: the CS tutoring center (second floor, about 2–6 PM, Mon–Fri) or Matt's office hours (Mon and Wed 9–11).

Appendix: Fixes

Find the fix number your step sent you to. Run the check, then the fix, then redo the step you came from.

#	You see	Fix
agate login		
F1	`Permission denied`, or it shows your laptop's name	Your command needs `your-unh-username@`. Retype it as `ssh your-unh-username@agate.cs.unh.edu`. A missing `@` is the usual typo.
F2	The password "doesn't type", or it's refused	Nothing showing is normal: type it blind and press Enter. Refused three times: switch to Route B and reset your password after class.
F3	You're not in the list, or have no account	Switch to Route B now. Tell Matt your username after class.
F4	It hangs, then `Connection timed out`	Join campus Wi-Fi (or the UNH VPN off campus) and try again. Still stuck: Route B.
Install and PATH		
F5	`opencode: command not found` after A4	Type `source ~/.bashrc`, then `opencode --version`. Still not found: type the four A4 lines again, one at a time.
F6	`EACCES` or `permission denied` during `npm install`	Don't use `sudo`. On agate, redo A4 from its first line. On a Mac, type these one at a time: `npm config set prefix ~/.npm-global` `echo 'export PATH="$HOME/.npm-global/bin:$PATH"' >> ~/.zshrc` Then close Terminal, open a new one, and redo B2.
F7	`node` not found, or `'opencode'` / `'npm'` "is not recognized" (Windows)	Node not installed: install the LTS version from nodejs.org, keeping "Add to PATH" ticked. Then close and reopen the terminal (a new window is needed after any install) and redo the step.
F8	"running scripts is disabled on this system" (Windows)	Type `npm.cmd` instead of `npm`, and `opencode.cmd` instead of `opencode`, for the rest of today.
F9	The version starts with `0.` or `2.`	Don't change it. Tell Matt the number, and do Part 2 on a neighbour's screen.
Python and the helper		
F10	`python3` not found, or the Microsoft Store opens (Windows)	On Windows, use `py -3` everywhere. If that isn't found either, install Python from python.org and reopen PowerShell.
F11	Mac asks to install "command line developer tools"	Click Install and wait. Then redo the step.
F12	`Existing OpenCode configuration differs`	Another settings file is already there. Don't move or delete it yourself: it may hold your own settings or a key. agate: if Step 0.3 said it's Monday's handout file, do the move in 0.3, then redo `prepare`. Anyone else: ask Matt.
F13	`An alternate OpenCode configuration exists`	Stop and don't change anything. Tell Matt, and do Part 2 on a neighbour's screen.
F14	`Key must be nonempty`	The paste didn't work. Copy the key again, run `key` again, and paste with right-click. Windows: if it now says the key folder already exists, see F21.

F15	`Piped input is refused`	You're inside an AI tool or Cursor. Run `key` in a normal Terminal or PowerShell window you opened yourself.
F16	`setup-key.py` isn't in Downloads, or the browser renamed it	Check the browser's downloads list. If it's called `setup-key (1).py`, use that exact name in quotes. If it opened as text instead of saving, right-click the page and choose "Save as".
Key and Deep Thought		
F17	`Configuration is invalid … bad file reference … api-key does not exist`	Your key isn't saved yet. Do the key step (A7 or B5).
F18	An error mentioning `401`, `403` or "unauthorized"	The key is wrong, or not allowed yet. Make a new key with Matt. agate and Mac: redo the key step. Windows: redoing the key step is refused once the key folder exists, so ask Matt (F21). Never show anyone the key.
F19	OpenCode spins with no reply for over a minute	Deep Thought may only answer on the UNH network: join campus Wi-Fi or the VPN, and try again. Still spinning? Raise your hand. Note the time and send it with Part 3; the cause may be the network or Deep Thought.
F20	A chat reply, but no `hello.py`	In OpenCode, type `/models` and choose DeepThought Claude Sonnet 4.6. Ask again, and allow it to create and run the file.
F21	`Windows key folder already exists` (after a failed first paste, or when replacing a key)	Don't delete anything. Raise your hand: Matt helps you set the folder aside, then you redo B5. Meanwhile do Part 2 on a neighbour's screen.
F22	"model not found" or "unsupported"	Don't switch to another provider. Write down the OpenCode version and the error, and send them with Part 3.